

Project Title	LLM Cluster — Free, Private AI for NYIT Students via Distributed Inference
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Faculty Advisor	Professor Austin Stietzel — Computer Science
Project Period	September 2026 — Spring 2027

Objective

Commercial AI tools cost \$20+/month per user, putting reliable AI access out of reach for many students. Free tiers are rate-limited and route private coursework through third parties.

LLM Cluster is a distributed inference platform that runs open-source large language models (Llama 3, Mistral) across a small fleet of dedicated worker machines hosted by the project. Students access the service through a lightweight client app — they never install or host worker software themselves. The system gives the NYIT community a free, private AI tool whose data never leaves project-controlled hardware.

The grant funds the research needed to turn an early proof-of-concept into a production-ready, evaluated system: scheduling under load, model selection, security, and a formal usability study.

Timeline and Deliverables

First Semester (14 weeks)

Weeks 1–3 — Literature review and benchmarking. Survey distributed inference systems and benchmark commercial baselines on a fixed prompt suite for latency and answer quality.

Deliverable: literature review + baseline dataset.

Weeks 4–8 — Core development. Design and implement load-balanced request routing, streaming responses, worker authentication, and an admin dashboard.

Add support for multiple model families.

Deliverable: working v1 system.

Weeks 9–14 — Testing and recalibration. Internal load testing (50 / 100 / 200 simulated concurrent users). Tune the scheduler. Begin desktop client refinement.

Deliverable: performance report + scheduler recommendation.

Second Semester (14 weeks)

Weeks 1–6 — Human subject study (IRB-approved). Recruit ~30 NYIT students. Within-subject comparison of LLM Cluster against a commercial baseline on real coursework tasks. Collect latency, answer-quality ratings, and trust measures. Begin mobile (iOS/Android) client development in parallel.

Deliverable: study dataset + mobile client beta.

Weeks 7–14 — Analysis, write-up, dissemination. Statistical analysis. Draft conference paper for an undergraduate research venue. Public release of the system and mobile app. Final presentation to CoECS faculty.

Deliverable: paper + public release + final report.

Signatures

Student A — Saba Karazanashvili _____ Date: _____

Student B — Ameer Teyah _____ Date: _____

Student C — Mate Milorava _____ Date: _____

Faculty Advisor — Prof. Austin Stietzel _____ Date: _____